

Arc Length and Improper Integrals



Arc length

- 1 Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.
 - 2 Set up (do not evaluate) the arc length integral for $y = \sin x$ on $[0, \pi]$.
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Improper integrals

- 3 Evaluate $\int_1^{\infty} \frac{1}{x^2} dx$, or state that it diverges.
 - 4 Evaluate $\int_1^{\infty} \frac{1}{x} dx$, or state that it diverges.
 - 5 Evaluate $\int_0^1 \frac{1}{\sqrt{x}} dx$ (improper at the lower limit).
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More arc length practice

- 6 Find the arc length of the line $y = 3x + 1$ from $x = 0$ to $x = 4$, and verify your answer using the distance formula.
 - 7 Set up (do not evaluate) the arc length integral for $y = \ln x$ on $[1, e]$.
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More improper integrals

- 8 Evaluate $\int_1^{\infty} \frac{1}{x^{1.5}} dx$, or state that it diverges.
 - 9 Evaluate $\int_{-\infty}^0 e^x dx$, or state that it diverges.
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The p-integral test

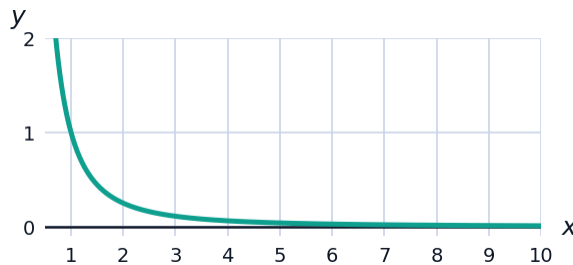
- 10 For $\int_1^{\infty} \frac{1}{x^p} dx$, state the condition on p for convergence, and explain why using the antiderivative $\frac{x^{1-p}}{1-p}$ (for $p \neq 1$).
 - 11 Use the p -integral result to quickly determine whether $\int_1^{\infty} \frac{1}{x^{0.9}} dx$ converges.
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Improper integrals with infinite discontinuities inside the interval

- 12 Evaluate $\int_0^2 \frac{1}{(x-1)^2} dx$, or explain why it diverges (note the integrand is undefined at $x = 1$, inside the interval).
- 13 Evaluate $\int_{-1}^1 \frac{1}{x^{1/3}} dx$ (an odd function with a discontinuity at $x = 0$), splitting at $x = 0$.
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Visualizing an improper integral as a limiting area

- 14 The graph shows $f(x) = \frac{1}{x^2}$ on $[1, 10]$. As the right boundary extends to infinity, the shaded area approaches a finite limit. Explain why this is plausible, using the fact that $\int_1^\infty \frac{1}{x^2} dx = 1$ (found earlier).



Arc length of a more involved curve

- 15 Set up (do not evaluate) the arc length integral for $y = x^2$ on $[0, 2]$, and estimate it using the straight-line distance between the endpoints as a lower bound.
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Comparison test for improper integrals

- 16 Use a comparison with $\int_1^\infty \frac{1}{x^2} dx$ (known to converge) to argue that $\int_1^\infty \frac{1}{x^2 + 1} dx$ also converges, without evaluating it.
- 17 Use a comparison with $\int_1^\infty \frac{1}{x} dx$ (known to diverge) to argue that $\int_1^\infty \frac{1}{\sqrt{x}} dx$ also diverges.
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A capstone arc-length application

- 18 A cable hangs in the shape $y = \frac{x^2}{8}$ between the towers at $x = -4$ and $x = 4$. Set up the integral for the length of the cable (do not evaluate).
- 19 Evaluate $\int_2^\infty \frac{1}{(x-1)^3} dx$, or state that it diverges.

- 20** A city's water tower is shaped by revolving the curve $y = \sqrt{25 - x^2}$ (the upper half of a circle of radius 5) about the x -axis; set up (do not evaluate) the arc length integral for this curve on $[-4, 4]$, which describes the profile of one side of the tower.